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1. ① 构建最大堆

$\{25, 13, 20, 8, 7, 17, 2, 5, 4\}$

② 得到第一个 25, 移至末尾后重新构建最大堆得

$\{20, 13, 17, 8, 7, 4, 2, 5, 25\}$

③ $\{17, 13, 5, 8, 7, 4, 2, 20, 25\}$

④ $\{13, 8, 5, 2, 7, 4, 17, 20, 25\}$

⑤ $\{8, 7, 5, 2, 4, 13, 17, 20, 25\}$

⑥ $\{7, 4, 5, 2, 8, 13, 17, 20, 25\}$

⑦ $\{5, 4, 2, 7, 8, 13, 17, 20, 25\}$

⑧ $\{4, 2, 5, 7, 8, 13, 17, 20, 25\}$

⑨ $\{2, 4, 5, 7, 8, 13, 17, 20, 25\}$

2. Heap-sort (A):

Create-min-heap(A)

for - in range(n)

B.insert(A[i])

if (A[i].next) A[i] = A[i].next

creat-min-heap(A) Else remove A[i]

A为用链表头结点构成的列表

B为最终排序好的链表

creat-min-heap(A)

构建最小堆

复杂度分析: 共 n 轮, 每轮为 $\log k \Rightarrow T(n) = O(n \log k)$



3. 由于 $0 < a \leq \frac{1}{2}$

$$\therefore 1-a \geq a$$

$\therefore$ 叶子结点最小深度应为一半为 a 比例的部分
最大深度则相反

设深度为 m

$$\therefore n \cdot a^m = 1 \Rightarrow m = -\log_a n \Rightarrow \text{最小深度为 } -\frac{\lg n}{\lg a}$$

$$n(1-a)^m = 1 \Rightarrow m = -\log_{(1-a)} n \Rightarrow \text{最大深度为 } -\frac{\lg n}{\lg(1-a)}$$

$$4. \textcircled{1} P_i = \frac{C_{i-1}' \cdot C_{n-i}'}{C_n^3} = \frac{b(i-1)(n-i)}{n(n-1)(n-2)}$$

$$\textcircled{2} P_{\lfloor \frac{n+1}{2} \rfloor} = \frac{3(n-1)^2}{2n(n-1)(n-2)} = \frac{3(n-1)}{2n(n-2)} \quad (\text{设 } n \text{ 为奇数, } n \text{ 的奇偶对本题无影响})$$

$$\therefore \frac{P_{\lfloor \frac{n+1}{2} \rfloor}}{P_0} = \frac{\frac{3(n-1)}{2n(n-2)}}{\frac{1}{n}} = \frac{3(n-1)}{2(n-2)} \rightarrow \frac{3}{2} \quad (n \rightarrow \infty)$$

$\therefore$ 变为之前的 $\frac{3}{2}$ 倍

$$\textcircled{3} \sum_{i=\frac{n}{3}}^{\frac{2}{3}n} P_i \approx \int_{\frac{n}{3}}^{\frac{2}{3}n} \frac{b(i-1)(n-i)}{n(n-1)(n-2)} di = \frac{b}{n(n-1)(n-2)} \int_{\frac{n}{3}}^{\frac{2}{3}n} [-i^2 + (n+1)i - n^2] di$$
$$= \frac{13n^2 - 27n}{27(n-1)(n-2)}$$

$$\therefore \frac{\sum_{i=\frac{n}{3}}^{\frac{2}{3}n} P_i}{\frac{1}{3}} = \frac{13n(n-\frac{27}{13})}{9(n-1)(n-2)} \rightarrow \frac{13}{9} \quad (n \rightarrow \infty)$$

$\textcircled{4}$ 快速排序最优复杂度为 $\Theta(n \lg n)$, 因此该方法肯定为 $\Omega(n \lg n)$

